# Appendix F: Material Properties

## F.1) Composite Material Properties

(These properties will be updated periodically during and after material coupon testing)

### Carbon/Epoxy Properties

Reference: Daniel, I. M. and O. Oshai, *Engineering Mechanics of Composite Materials*, 2nd Edition, 2006, page 380.

| *Type* | *Symbol* | Unidirectional | Woven (5-Harness) | *Units* |
|---|---|---|---|---|
| *Carbon Fiber* | | AS4 | AGP370-5H | |
| *Epoxy Resin* | | 3501-6 | 3501-6S | |
| *Update* | | 2/17/2018 | 2/17/2018 | |
| *Fiber Volume* | $V_f$ | 63 | 62 | *%* |
| *Density* | $\rho$ | 0.058 | 0.058 | *lb/in$^3$* |
| *Modulus, Longitudinal* | $E_{11}$ | 21.3 | 11.2 | *Msi* |
| *Modulus, Transverse, In-Plane* | $E_{22}$ | 1.50 | 10.9 | *Msi* |
| *Modulus, Transverse, Out-of-Plane* | $E_{33}$ | 1.50 | 2.0 | *Msi* |
| *Shear Modulus, In-Plane* | $G_{12}$ | 1.00 | 0.94 | *Msi* |
| *Shear Modulus, Out-of-Plane* | $G_{23}$ | 0.54 | 0.59 | *Msi* |
| *Shear Modulus, Out-of-Plane* | $G_{13}$ | 1.00 | 0.74 | *Msi* |
| *Poisson Ratio, In-Plane* | $\nu_{12}$ | 0.27 | 0.06 | - |
| *Poisson Ratio, Out-of-Plane* | $\nu_{23}$ | 0.47 | 0.37 | - |
| *Poisson Ratio, Out-of-Plane* | $\nu_{13}$ | 0.27 | 0.50 | - |
| *Tension Strength, Longitudinal* | $F_{1T}$ | 330. | 140 | *Ksi* |
| *Tension Strength, Transverse* | $F_{2T}$ | 8.30 | 124 | *Ksi* |
| *Tension Strength, Out-of-Plane* | $F_{3T}$ | 8.30 | 8.70 | *Ksi* |
| *Compression Strength, Longitudinal* | $F_{1C}$ | -250. | -130. | *Ksi* |
| *Compression Strength, Transverse* | $F_{2C}$ | -33.0 | -130. | *Ksi* |
| *Compression Strength, Out-of-Plane* | $F_{3C}$ | -33.0 | -118. | *Ksi* |
| *Shear Strength, In-Plane* | $F_{S6}$ | 11.0 | 10.3 | *Ksi* |
| *Shear Strength, Out-of-Plane* | $F_{S4}$ | - | 9.50 | *Ksi* |
| *Shear Strength, Out-of-Plane* | $F_{S5}$ | - | 10.8 | *Ksi* |
| *Thermal Expansion, Longitudinal* | $\alpha_{11}$ | -0.50 | 1.90 | *10$^{-6}$/$^o$F* |
| *Thermal Expansion, Transverse* | $\alpha_{22}$ | 15.0 | 2.10 | *10$^{-6}$/$^o$F* |
| *Thermal Expansion, Transverse* | $\alpha_{33}$ | 15.0 | 29.0 | *10$^{-6}$/$^o$F* |
| *Moisture Expansion, Longitudinal* | $\beta_{11}$ | -0.50 | 1.90 | *10$^{-6}$/%* |
| *Moisture Expansion, Transverse* | $\beta_{22}$ | 15.0 | 2.10 | *10$^{-6}$/%* |
| *Moisture Expansion, Transverse* | $\beta_{33}$ | 15.0 | 29.0 | *10$^{-6}$/%* |
| *Ply Thickness* | $t_{ply}$ | TBD | TBD | *Inch* |

## Glass/Epoxy Properties

References: (1) Daniel, I. M. and O. Oshai, *Engineering Mechanics of Composite Materials*, 2nd Edition, 2006, page 380. (2) Tomblin, J., McKenna, J., Ng, Y., and Raju, K. S., "B-Basis Design Allowables for Epoxy-Based Prepreg; FiberCote E-Glass Fabric 7781/E765," Advanced General Aviation Transport Experiments, Report AGATE-WP3.3-033051-105, 2001, pg. 217

| *Type* | *Symbol* | Unidirectional | Woven (8-Harness) | | *Units* |
|---|---|---|---|---|---|
| *Glass Fiber* | | E-glass | M10E | *7781* | |
| *Epoxy Resin* | | 3783 | 3783 | *E765* | |
| *Update* | | | 2/17/2018 | *1/31/2020* | |
| *Fiber Volume* | $V_f$ | 55 | 50 | *45-49* | *%* |
| *Density* | $\rho$ | .071 | 0.068 | *0.08* | *lb/in$^3$* |
| *Modulus, Longitudinal* | $E_{11}$ | 6 | 3.55 | *3.73* | *Msi* |
| *Modulus, Transverse, In-Plane* | $E_{22}$ | 1.5 | 3.45 | *3.52* | *Msi* |
| *Modulus, Transverse, Out-of-Plane* | $E_{33}$ | 1.5 | 1.68 | *1.5* | *Msi* |
| *Shear Modulus, In-Plane* | $G_{12}$ | 0.62 | 0.68 | *0.70* | *Msi* |
| *Shear Modulus, Out-of-Plane* | $G_{23}$ | 0.50 | 0.52 | | *Msi* |
| *Shear Modulus, Out-of-Plane* | $G_{13}$ | 0.62 | 0.38 | | *Msi* |
| *Poisson Ratio, In-Plane* | $\nu_{12}$ | 0.28 | 0.11 | *0.157* | - |
| *Poisson Ratio, Out-of-Plane* | $\nu_{23}$ | 0.50 | 0.20 | | - |
| *Poisson Ratio, Out-of-Plane* | $\nu_{13}$ | 0.28 | 0.15 | | - |
| *Tension Strength, Longitudinal* | $F_{1T}$ | 165 | 62.8 | *60.8* | *Ksi* |
| *Tension Strength, Transverse* | $F_{2T}$ | 5.7 | 55.9 | *47.3* | *Ksi* |
| *Tension Strength, Out-of-Plane* | $F_{3T}$ | 5.7 | 3.9 | | *Ksi* |
| *Compression Strength, Longitudinal* | $F_{1C}$ | -90 | -54.6 | *-67.9* | *Ksi* |
| *Compression Strength, Transverse* | $F_{2C}$ | -18.6 | -48.6 | *-51.3* | *Ksi* |
| *Compression Strength, Out-of-Plane* | $F_{3C}$ | -18.6 | -34.4 | | *Ksi* |
| *Shear Strength, In-Plane* | $F_{S6}$ | 12.9 | 12.2 | *17.4* | *Ksi* |
| *Shear Strength, Out-of-Plane* | $F_{S4}$ | | 6.3 | *6.59* | *Ksi* |
| *Shear Strength, Out-of-Plane* | $F_{S5}$ | | 5.9 | | *Ksi* |
| *Thermal Expansion, Longitudinal* | $\alpha_{11}$ | 3.9 | | *5.6* | *$10^{-6}$/°F* |
| *Thermal Expansion, Transverse* | $\alpha_{22}$ | 14.4 | | *5.6* | *$10^{-6}$/°F* |
| *Thermal Expansion, Transverse* | $\alpha_{33}$ | 14.4 | | | *$10^{-6}$/°F* |
| *Moisture Expansion, Longitudinal* | $\beta_{11}$ | 0.0 | | | *$10^{-6}$/%* |
| *Moisture Expansion, Transverse* | $\beta_{22}$ | 0.2 | | *0.06* | *$10^{-6}$/%* |
| *Moisture Expansion, Transverse* | $\beta_{33}$ | 0.2 | | *0.06* | *$10^{-6}$/%* |
| *Ply Thickness* | $t_{ply}$ | TBD | TBD | *0.0085* | *Inch* |

## F.2) Carbon/Epoxy Properties – Unidirectional Tape [ Wet or VaRTM ]

| | Carbon/Epoxy Unidirectional Tape, 4.3 ounce/square yard | | | |
|---|---|---|---|---|
| *Fabrication* | Wet or VaRTM | Wet | | |
| *Update* | Nominal | 3/13/2017 | | |
| $V_f$ (%) | 56.73 | 48.7 | | |
| $E_L$ (Msi) | 19.3 | 16.22 | | |
| $E_T$ (Msi) | 1.35 | 0.9 | | |
| $G_{LT}$ (Msi) | 0.623 | 0.1389 | | |
| $\nu_{LT}$ | 0.269 | 0.2687 | | |
| $F_{1T}$ (Ksi) | 210 | 141.5 | | |
| $F_{1C}$ (Ksi) | -157 | -111 | | |
| $F_{2T}$ (Ksi) | 7.02 | 2.76 | | |
| $F_{2C}$ (Ksi) | -14.91 | -5.04 | | |
| $F_s$ (Ksi) | 10.51 | 6.06 | | |
| $\rho$ (lb/in$^3$) | 0.0564 | 0.05460 | | |
| $t_{ply}$ (inch) | 0.005 | 0.007 | | |

## F.3) Carbon/Epoxy Properties – Woven Fabric [ Wet or VaRTM ]

| | Carbon/Epoxy Woven (Plain Weave) Fabric, 5.7 ounce/square yard | | | |
|---|---|---|---|---|
| *Fabrication* | Wet or VaRTM | Wet | VaRTM | |
| *Update* | Nominal | 3/13/2017 | 3/13/2017 | |
| $V_f$ (%) | 0.5378 | 0.47 | .4108 | |
| $E_L$ (Msi) | 4.5 | 8.52 | 7.51 | |
| $E_T$ (Msi) | 4.5 | 8.52 | 7.54 | |
| $G_{LT}$ (Msi) | 0.699 | .59 | .524 | |
| $\nu_{LT}$ | 0.1165 | .038 | .0412 | |
| $F_{1T}$ (Ksi) | 84 | 96.4 | 93.16 | |
| $F_{1C}$ (Ksi) | -66 | -93.6 | -88.4 | |
| $F_{2T}$ (Ksi) | 84 | 96.4 | 93.16 | |
| $F_{2C}$ (Ksi) | -66 | -93.6 | -88.4 | |
| $F_s$ (Ksi) | 13.1 | 14.35 | 14.1 | |
| $\rho$ (lb/in$^3$) | 0.0768 | .0473 | .0471 | |
| $t_{ply}$ (inch) | 0.009 | 0.0083 | 0.0083 | |

## F.4) Carbon/Epoxy Properties – Unidirectional Tape [ Prepreg, Vacuum Oven ]

| | Carbon/Epoxy Unidirectional tape, M55J/EX1515 | | | |
|---|---|---|---|---|
| *Update* | Nominal | | | |
| $V_f$ (%) | | | | |
| $E_L$ (Msi) | 38 | | | |
| $E_T$ (Msi) | 1.185 | | | |
| $G_{LT}$ (Msi) | 0.434 | | | |
| $\nu_{LT}$ | 0.283 | | | |
| $F_{1T}$ (Ksi) | 253 | | | |
| $F_{1C}$ (Ksi) | -90 | | | |
| $F_{2T}$ (Ksi) | 12.17 | | | |
| $F_{2C}$ (Ksi) | -16.22 | | | |
| $F_s$ (Ksi) | 8.59 | | | |
| $\rho$ (lb/in$^3$) | 0.0549 | | | |
| $t_{ply}$ (inch) | 0.0031 | | | |

## F.5) Carbon/Epoxy Properties – Woven Fabric [ Prepreg, Vacuum Oven ]

| | Carbon/Epoxy Woven (2x2 Twill) Fabric CF0302/MTM49LT | | | |
|---|---|---|---|---|
| *Update* | Nominal | | | |
| $V_f$ (%) | .567 | | | |
| $E_L$ (Msi) | 10.28 | | | |
| $E_T$ (Msi) | 10.28 | | | |
| $G_{LT}$ (Msi) | .623 | | | |
| $\nu_{LT}$ | .0316 | | | |
| $F_{1T}$ (Ksi) | 106.0 | | | |
| $F_{1C}$ (Ksi) | -87300 | | | |
| $F_{2T}$ (Ksi) | 106.0 | | | |
| $F_{2C}$ (Ksi) | -87300 | | | |
| $F_s$ (Ksi) | 10,500 | | | |
| $\rho$ (lb/in$^3$) | .0564 | | | |
| $t_{ply}$ (inch) | 0.006 | | | |

## F.6) Glass/Epoxy Properties – Unidirectional Tape [ Wet or VaRTM process ]

| | Glass/Epoxy Unidirectional Tape | | | |
|---|---|---|---|---|
| *Fabrication* | Wet or VaRTM | Wet | | |
| *Update* | Nominal | 3/13/2017 | | |
| $V_f$ (%) | 50.0 | 52.7 | | |
| $E_L$ (Msi) | 5.29 | 6.76 | | |
| $E_T$ (Msi) | 0.408 | 1.72 | | |
| $G_{LT}$ (Msi) | 0.148 | 0.638 | | |
| $\nu_{LT}$ | 0.265 | 0.258 | | |
| $F_{1T}$ (Ksi) | 156.0 | 104. | | |
| $F_{1C}$ (Ksi) | -105 | -53.37 | | |
| $F_{2T}$ (Ksi) | 2.69 | 12.45 | | |
| $F_{2C}$ (Ksi) | -5.00 | -25.6 | | |
| $F_s$ (Ksi) | 6.07 | 5.67 | | |
| $\rho$ (lb/in$^3$) | 0.700 | 0.0677 | | |
| $t_{ply}$ (inch) | .009 | 0.012 | | |

## F.7) Glass/Epoxy Properties – Woven Fabric [ Wet or VaRTM process ]

| | E-Glass/Epoxy woven (plain weave) fabric, 9.5 ounce/sq. yard | | | |
|---|---|---|---|---|
| *Fabrication* | Wet or VaRTM | Wet | VaRTM | |
| *Update* | Nominal | 3/13/2017 | 3/13/2017 | |
| $V_f$ (%) | 0.5378 | .428 | .459 | |
| $E_L$ (Msi) | 4.5 | 3.24 | 3.35 | |
| $E_T$ (Msi) | 4.5 | 3.53 | 3.35 | |
| $G_{LT}$ (Msi) | 0.67 | .5219 | .558 | |
| $\nu_{LT}$ | 0.116 | .1306 | .1287 | |
| $F_{1T}$ (Ksi) | 84 | 46.17 | 55.48 | |
| $F_{1C}$ (Ksi) | -66 | -30.9 | | |
| $F_{2T}$ (Ksi) | 84 | 48.84 | 55.42 | |
| $F_{2C}$ (Ksi) | -66 | -30.9 | -31.0 | |
| $F_s$ (Ksi) | 13.1 | 9.92 | 10.0 | |
| $\rho$ (lb/in$^3$) | 0.0768 | 0.0627 | 0.0641 | |
| $t_{ply}$ (inch) | 0.009 | 0.010 | 0.010 | |

## F.8) Core Material Properties

(These properties will be updated periodically during and after material coupon testing)

| ***Brand*** | **Nomex** | **Nomex** | **Divinycell** | **Divinycell** | **Baltek** |
|---|---|---|---|---|---|
| ***Type*** | **Honeycomb** | **Honeycomb** | **Foam** | **Foam** | **Balsa** |
| ***Material*** | **Aramid** | **Aluminum** | **H-45** | **H-250** | **Wood** |
| ***Model*** | **HRH10-1/8-1.8** | **5052-1/8-0.0007** | **Low density** | **High density** | **8 lb/ft$^3$** |
| $E_C$ | 8 ksi | 75 ksi | 7.25 ksi | 58 ksi | 500 ksi |
| $F_C$ (min) | 85 psi | 200 psi | 72 psi | 885 psi | 1180 psi |
| $F_C$ (min-stable) | 95 psi | 215 psi | 72 psi | 885 psi | 1180 psi |
| $F_C$ (crush) | | 130 psi | 72 psi | 885 psi | 1690 psi |
| $G_L$ | 3.8 ksi | 45 ksi | 2.17 ksi | 14.1 ksi | 26.7 ksi |
| $F_{SL}$ (min) | 75 psi | 155 psi | 67 psi | 566 psi | 140 psi |
| $G_W$ | 1.5 ksi | 22 ksi | 2.17 ksi | 14.1 ksi | 18.3 ksi |
| $F_{SW}$ (min) | 40 psi | 90 psi | 67 psi | 566 psi | 140 psi |
| $\rho$ | 1.8 lb/ft$^3$ | 3.1 lb/ft$^3$ | 3.0 lb/ft$^3$ | 15.6 lb/ft$^3$ | 8 lb/ft$^3$ |
| *t (testing)** | 0.500 inch | 0.625 inch | 0.5 inch | 0.5 inch | 0.5 inch |

* Note: All of these cores are available in a wide variety of thicknesses.